EWB CLOUD LABS | STUDENT PROJECT REPORT

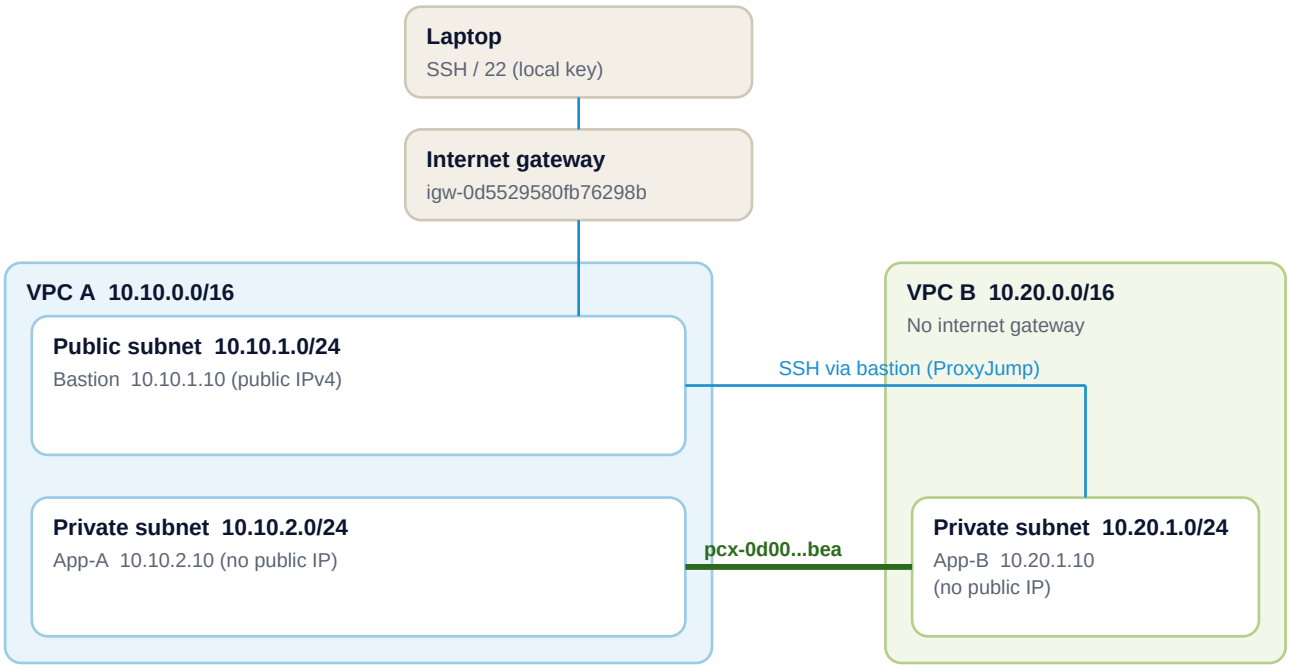
AWS VPC Peering

Connect two VPCs. Prove private connectivity.

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Batch	_____
Project	Project 01 - AWS VPC Peering (AWS Architect Associate)
Region	Asia Pacific (Mumbai) - ap-south-1
Completion date	28 October 2026

Scope: two VPCs (10.10.0.0/16 and 10.20.0.0/16) connected by a VPC peering connection, one public bastion in VPC A, and two private application instances. All tests use private IPv4 addresses.

1. Architecture and inventory



Administration path: laptop to bastion to App-A or App-B (SSH ProxyJump). Project traffic: App-A and App-B talk over their private IPs through the peering connection. VPC B has no internet gateway and the peering connection does not give it internet access.

Resource inventory (actual IDs)

Resource	Name tag	ID / address
VPC A	Meena-VPC-A	vpc-05d438aa13137e4b7 / 10.10.0.0/16
VPC B	Meena-VPC-B	vpc-01c402efa1b96c2ad / 10.20.0.0/16
Public subnet (A)	Meena-A-Public	10.10.1.0/24
Private subnet (A)	Meena-A-Private	10.10.2.0/24
Private subnet (B)	Meena-B-Private	10.20.1.0/24
Internet gateway	Meena-IGW-A	igw-0d5529580fb76298b (attached to VPC A)
Route tables	Meena-RT-A-Public Meena-RT-A-Private Meena-RT-B-Private	rtb-0dd241da78e9147f8 rtb-0df45ed2104373a0b rtb-0bf5f06b9d1649179
Peering connection	Meena-VPC-A-to-VPC-B	pcx-0d003244ac1299bea
Security groups	Meena-SG-Bastion Meena-SG-App-A Meena-SG-App-B	sg-0fce963c35fed0d33 sg-017f4fbe4b75a51a7 sg-002f14946ca58b19a
Instances	Meena-Bastion Meena-App-A Meena-App-B	i-09b0217a943965835 / 10.10.1.10 / public 13.232.179.239 i-018b89980d6b232fa / 10.10.2.10 / no public IP i-0630872fadcf72a60 / 10.20.1.10 / no public IP

2. Subnets, route tables and associations

Each subnet is explicitly associated with its own route table. The public subnet reaches the internet through the internet gateway; both private subnets only have the local route plus the peering route.

Route table	Destination	Target	Purpose
Meena-RT-A-Public (Meena-A-Public)	10.10.0.0/16 0.0.0.0/0 10.20.0.0/16	local igw-0d5529... pcx-0d00...	Bastion to App-A Bastion internet access Bastion SSH to App-B (ProxyJump)
Meena-RT-A-Private (Meena-A-Private)	10.10.0.0/16 10.20.0.0/16	local pcx-0d00...	Existing VPC A route App-A to VPC B (forward path)
Meena-RT-B-Private (Meena-B-Private)	10.20.0.0/16 10.10.0.0/16	local pcx-0d00...	Existing VPC B route Return path to VPC A

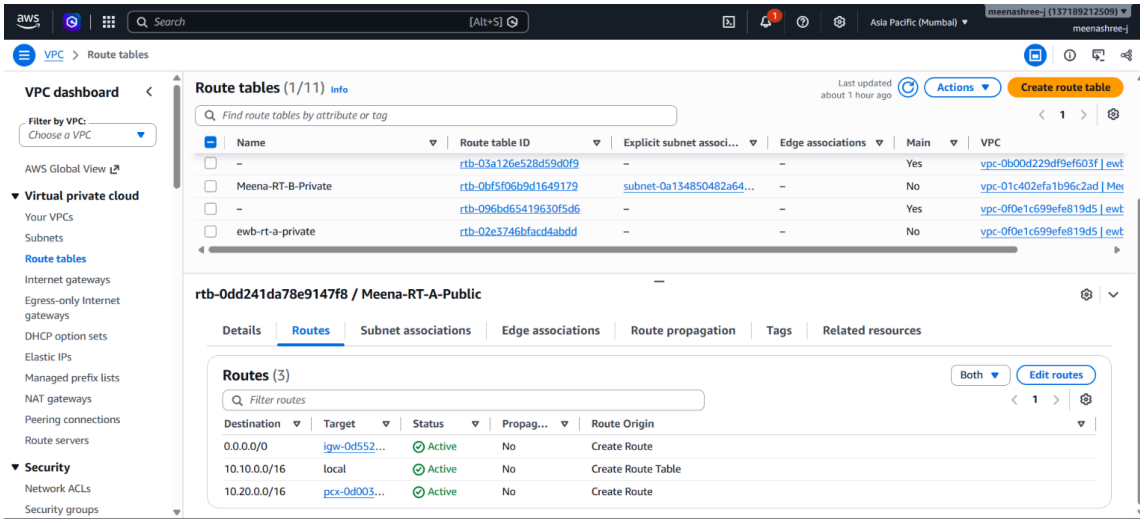


Figure 1. Meena-RT-A-Public - routes: local, 0.0.0.0/0 to IGW, 10.20.0.0/16 to peering. Result: all Active.

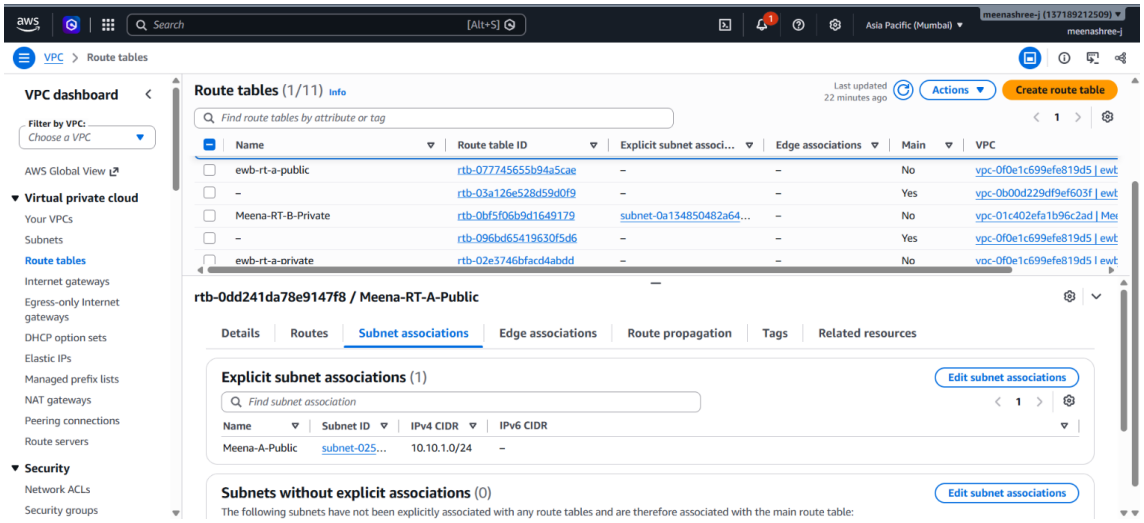


Figure 2. Meena-RT-A-Public - explicit association: Meena-A-Public (10.10.1.0/24).

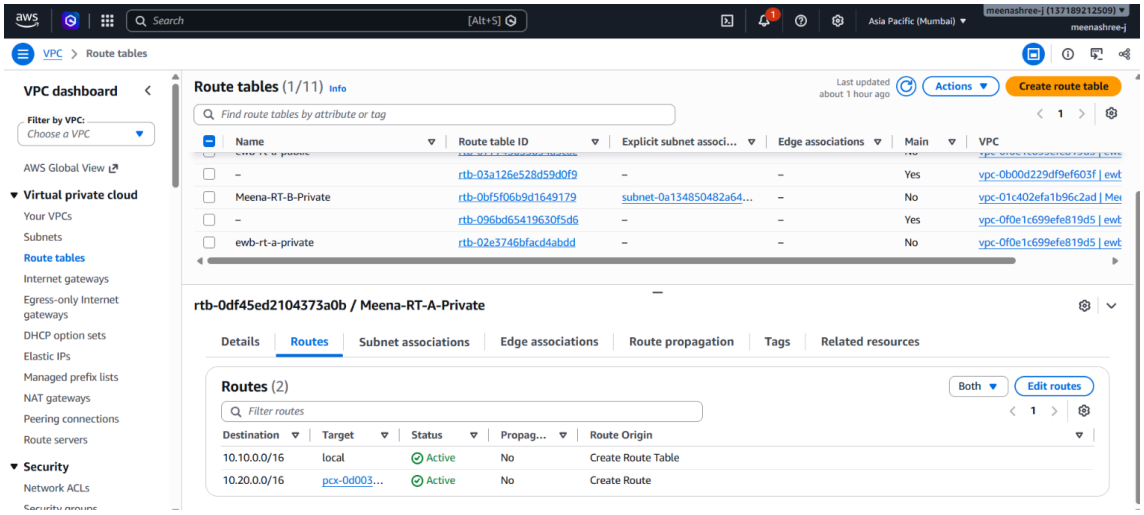


Figure 3. Meena-RT-A-Private - routes: local and 10.20.0.0/16 to peering (forward path). Result: Active.

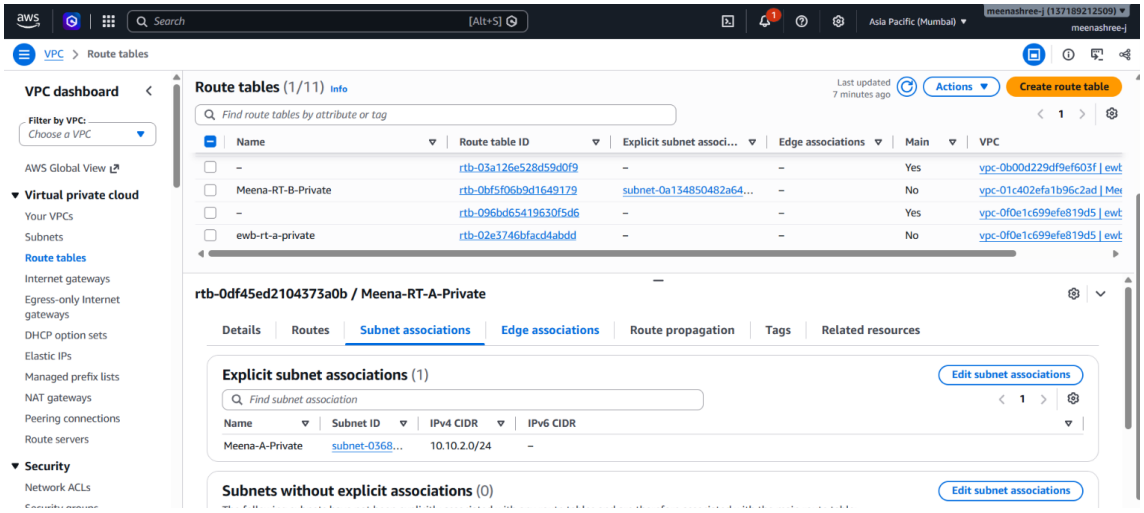


Figure 4. Meena-RT-A-Private - explicit association: Meena-A-Private (10.10.2.0/24).

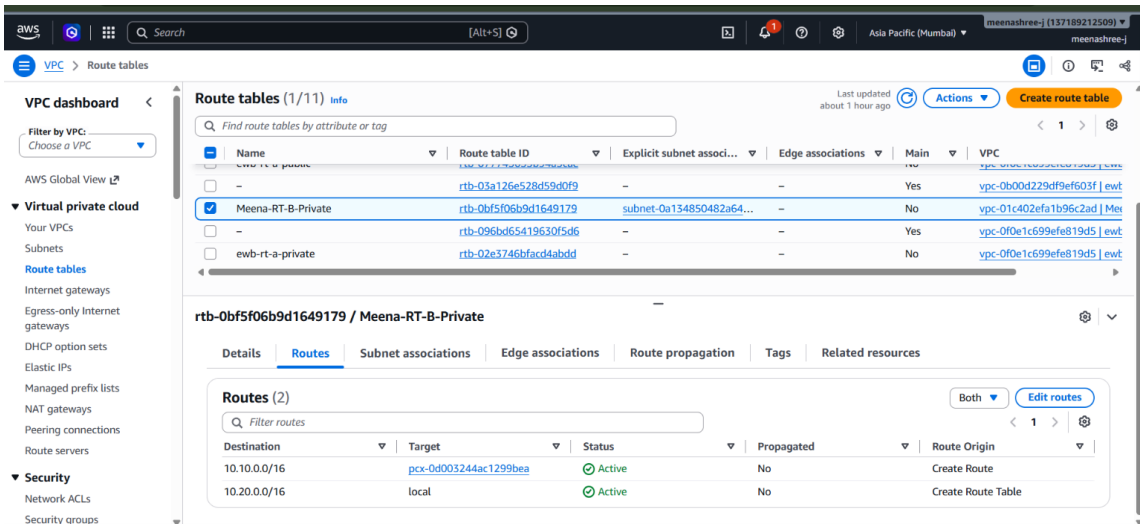


Figure 5. Meena-RT-B-Private - routes: local and 10.10.0.0/16 to peering (return path). Result: Active.

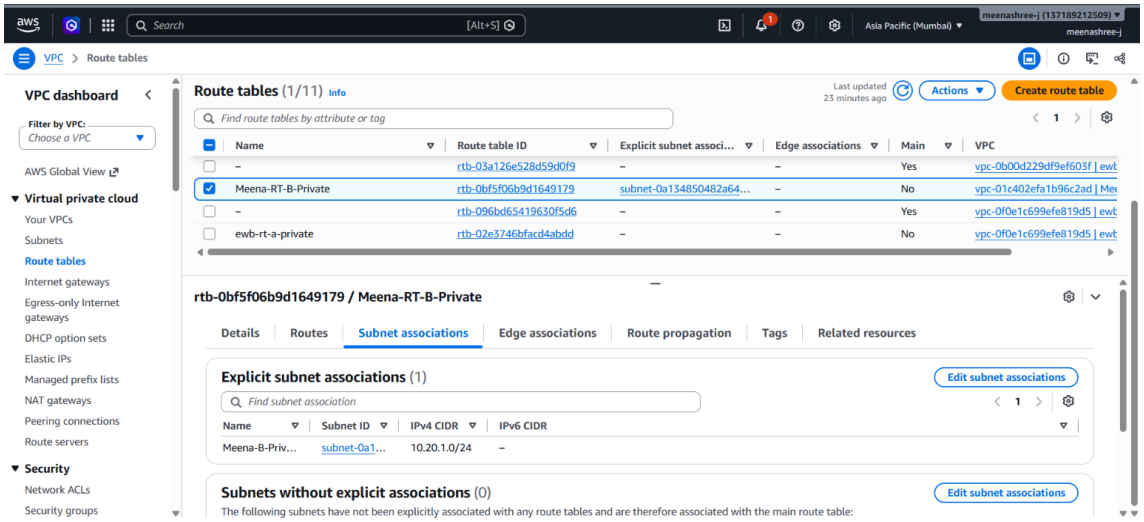


Figure 6. Meena-RT-B-Private - explicit association: Meena-B-Private (10.20.1.0/24).

3. Peering connection and security groups

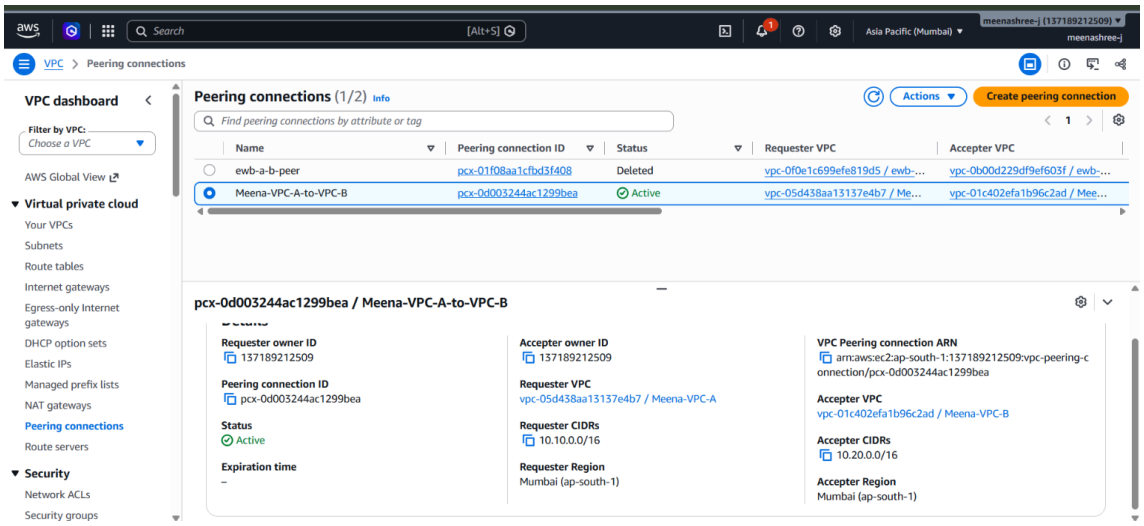


Figure 7. Peering connection pcx-0d003244ac1299bea - Status Active; requester VPC A (10.10.0.0/16), accepter VPC B (10.20.0.0/16), same account and Region.

Inbound rules (outbound left as default allow-all)

Security group	Rule	Source
Meena-SG-Bastion (VPC A)	SSH / TCP 22	My public IPv4 /32
Meena-SG-App-A (VPC A)	SSH / TCP 22	10.10.1.10/32 (bastion private IP)
Meena-SG-App-A (VPC A)	Custom ICMP IPv4 - Echo Request	10.20.1.10/32 (App-B)
Meena-SG-App-B (VPC B)	SSH / TCP 22	10.10.1.10/32 (bastion private IP)
Meena-SG-App-B (VPC B)	Custom ICMP IPv4 - Echo Request	10.10.2.10/32 (App-A)

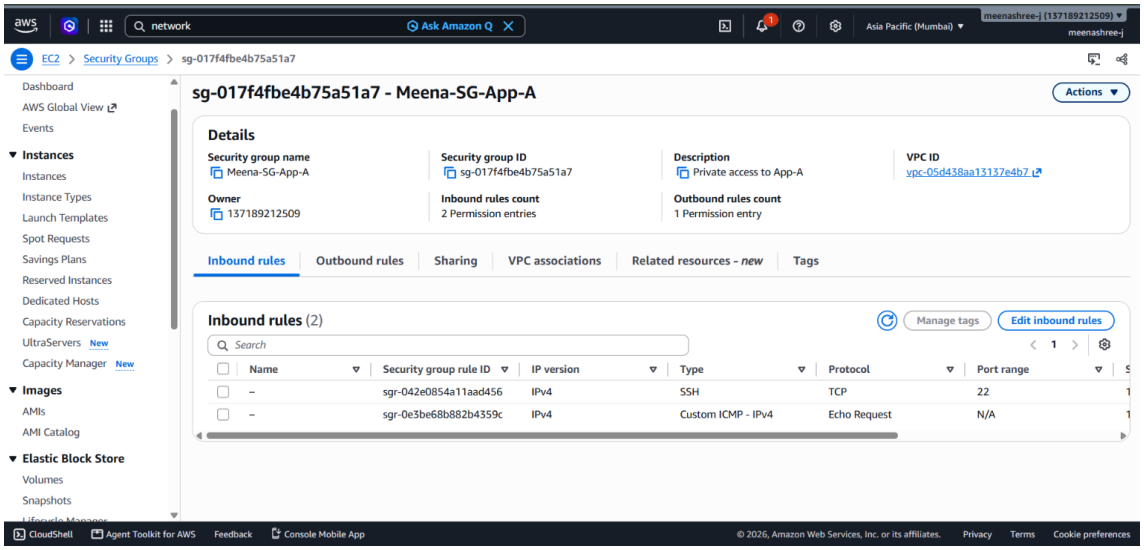


Figure 8. Meena-SG-App-A - inbound rules: SSH 22 and Custom ICMP IPv4 Echo Request.

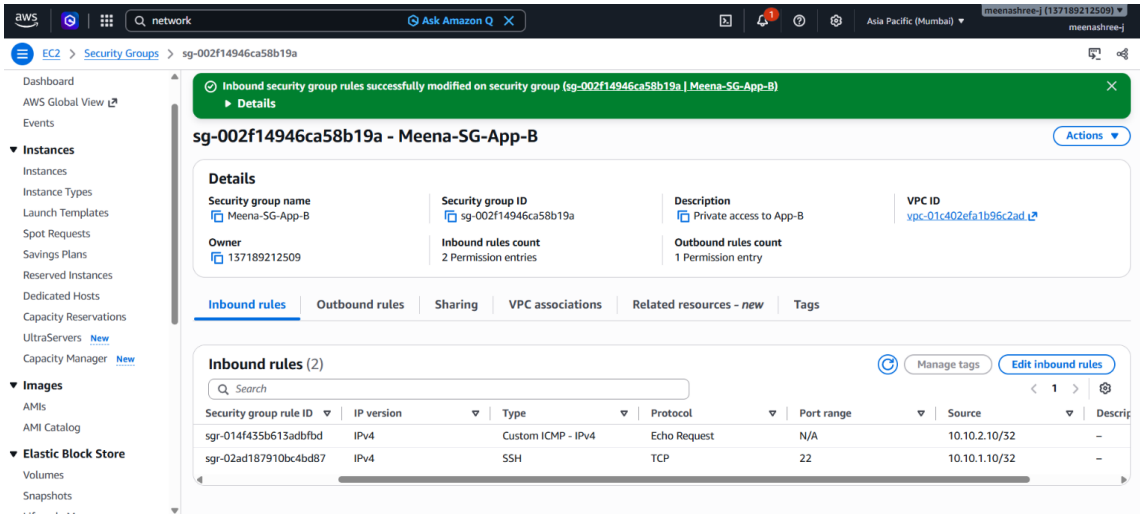


Figure 9. Meena-SG-App-B - inbound rules: Echo Request from 10.10.2.10/32 and SSH 22 from 10.10.1.10/32 (rule modified successfully).

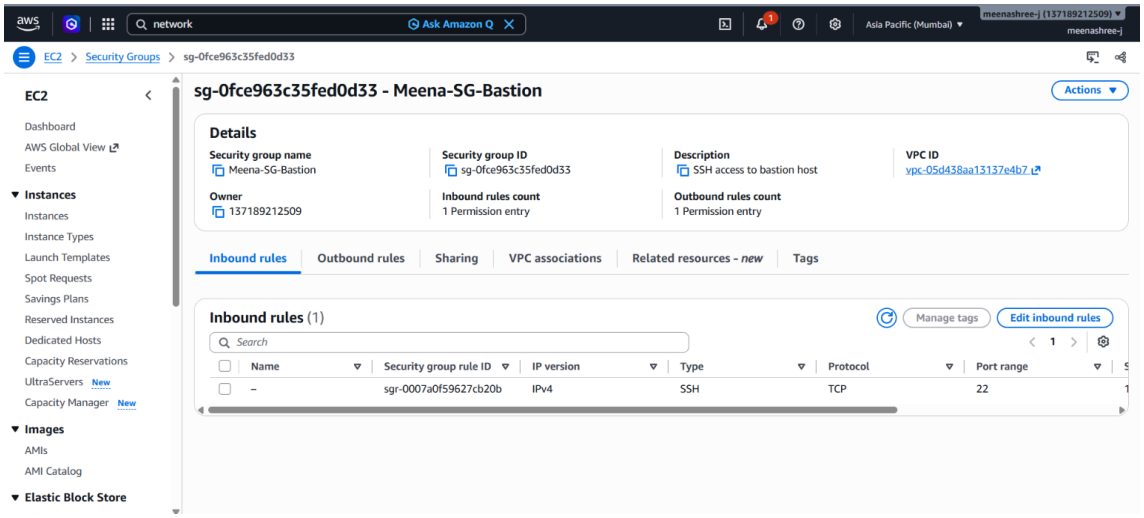


Figure 10. Meena-SG-Bastion - inbound rule: SSH / TCP 22 (source restricted to my public IP /32).

4. Test evidence

Test 01 - Baseline (no path)

To show the failure case, the peering route 10.20.0.0/16 was temporarily removed from Meena-RT-A-Private. The ping from App-A to App-B then failed. The route was restored afterwards and the ping worked again.



Figure 11. Host: App-A | Test: `ping -c 4 10.20.1.10` with peering route removed | Result: 4 transmitted, 0 received, 100% packet loss.

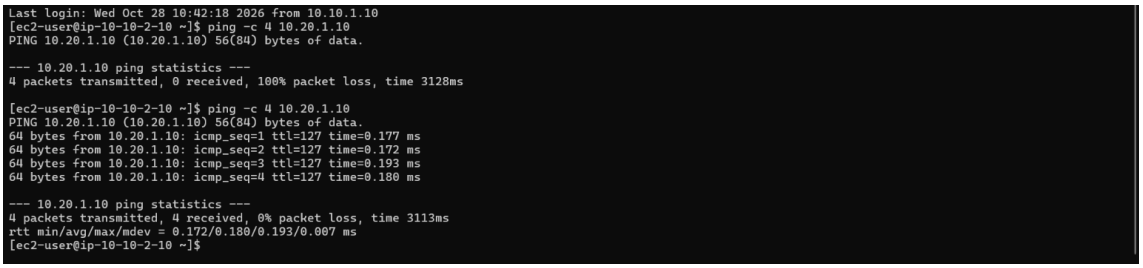


Figure 12. Host: App-A | Test: same ping, route restored | Result: 100% loss before, then 4 replies, 0% packet loss after the route was restored.

Tests 02 and 03 - Forward and reverse path

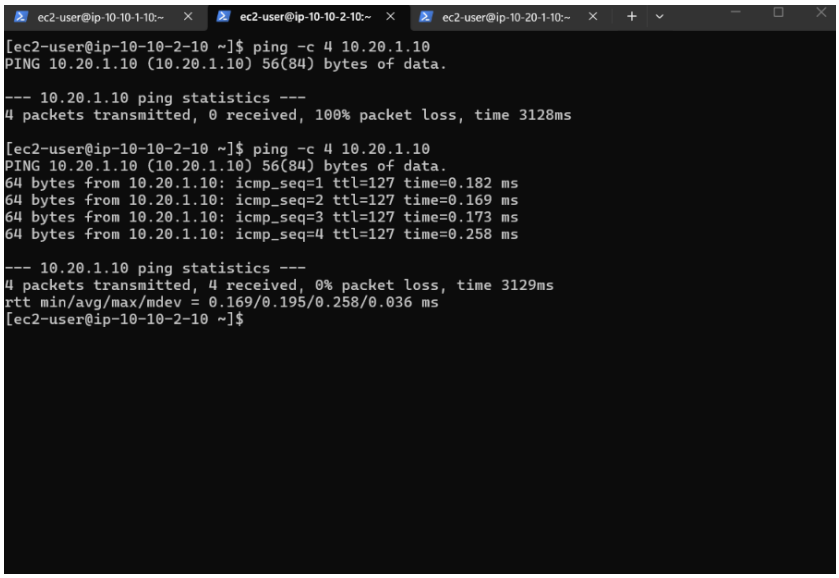


Figure 13. Host: App-A (10.10.2.10) | `ping -c 4 10.20.1.10` | Result: 4 replies, 0% packet loss.

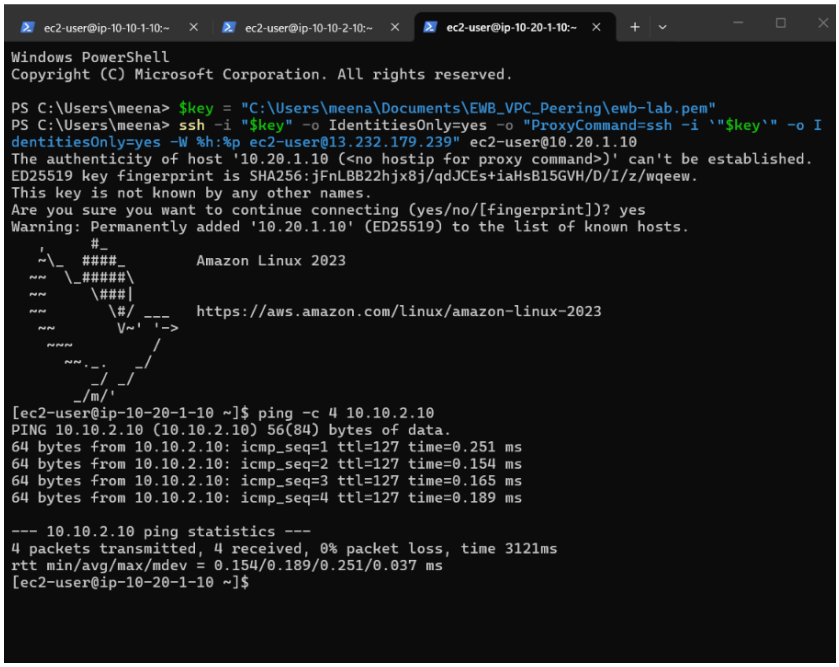


Figure 14. Host: App-B (10.20.1.10) | `ping -c 4 10.10.2.10` | Result: 4 replies, 0% packet loss. SSH reached App-B through the bastion.

Test 04 - Private SSH to App-B through the bastion

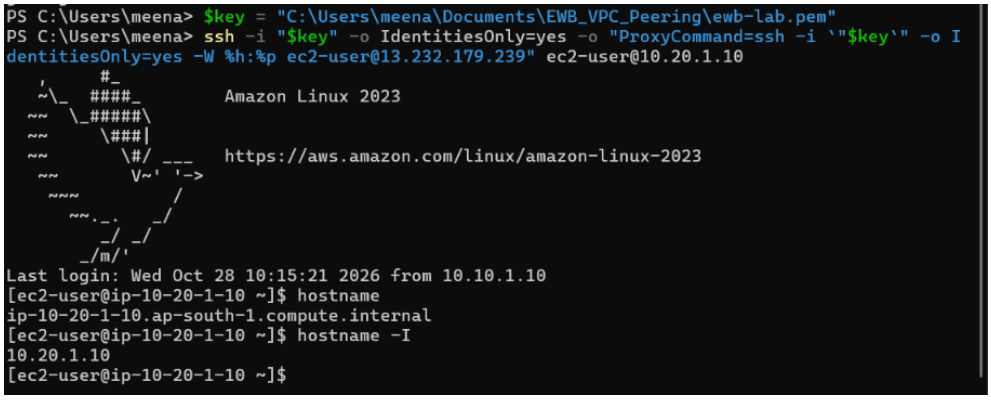


Figure 15. Host: laptop to App-B | SSH via ProxyJump through 13.232.179.239 | Result: shell on ip-10-20-1-10; `hostname -I` = 10.20.1.10; "Last login ... from 10.10.1.10" shows the connection came from the bastion private IP.

Test 05 - Isolation

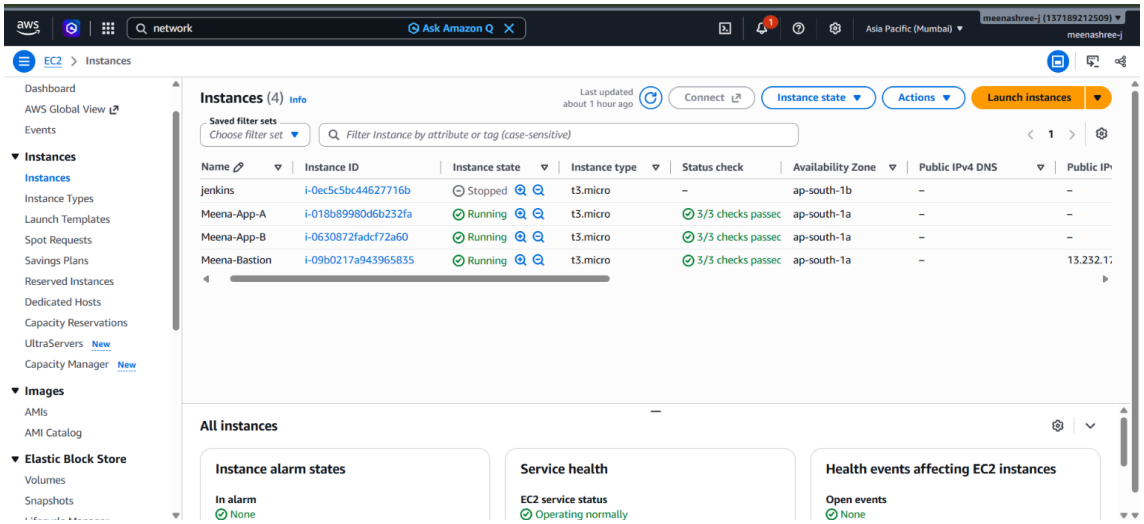


Figure 16. EC2 instances | Meena-App-A and Meena-App-B show no public IPv4 (-); only Meena-Bastion has a public address.

5. Controlled failure and recovery

Procedure: with pings working, the Echo Request rule (source 10.10.2.10/32) was removed from Meena-SG-App-B while the SSH rule stayed in place. The forward ping from App-A then failed. The exact same rule was added back and the ping recovered.

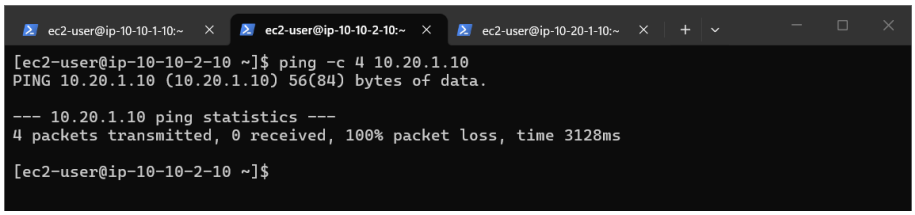


Figure 17. Host: App-A | `ping -c 4 10.20.1.10` with the App-B ICMP rule removed | Result: 100% packet loss.

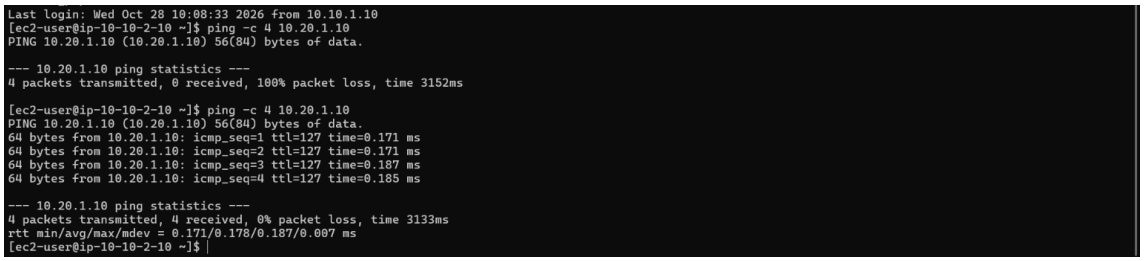


Figure 18. Host: App-A | same ping before and after the rule was restored | Result: 100% loss, then 4 replies, 0% packet loss.

Explanation

Routing and peering were unchanged, so the packets still had a path. Removing the Echo Request rule made the destination security group drop them, so the ping failed. Restoring the rule made the ping work again. This shows that an Active peering connection and correct routes are not enough on their own: the destination security group must also allow the traffic.

6. Forward path, return path and what I learned

Forward path (App-A to App-B)

App-A (10.10.2.10) sends a packet to 10.20.1.10. Its subnet route table, Meena-RT-A-Private, matches 10.20.0.0/16 and sends the packet to the peering connection. It arrives in VPC B, and Meena-SG-App-B allows ICMP Echo Request from 10.10.2.10/32.

Return path (App-B to App-A)

The reply is allowed automatically because security groups are stateful. It still needs a route, so Meena-RT-B-Private sends 10.10.0.0/16 to the peering connection. Without that route the request would arrive but the reply would never get back.

Bastion path to App-B

With ProxyJump the bastion opens the SSH connection to App-B, so the public subnet route table also needs 10.20.0.0/16 to the peering connection. App-B therefore sees the bastion private IP (10.10.1.10) as the SSH source, which is why its security group allows that address and not my laptop.

What I learned

Peering is only the link; route tables on both sides and security groups decide whether traffic actually flows. A timeout on SSH to the bastion pointed to my security group source or network, and not to the key. I changed the ICMP rules from all ICMP to Echo Request only, to allow just the required traffic. Missing routes and blocked rules both show up as silent packet loss, so I had to check route tables, associations and rules one at a time.